

Wednesday warm-up: Kinematics, II and III

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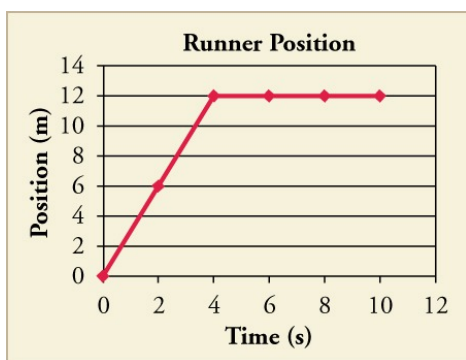


Figure 1: The position of a runner versus time.

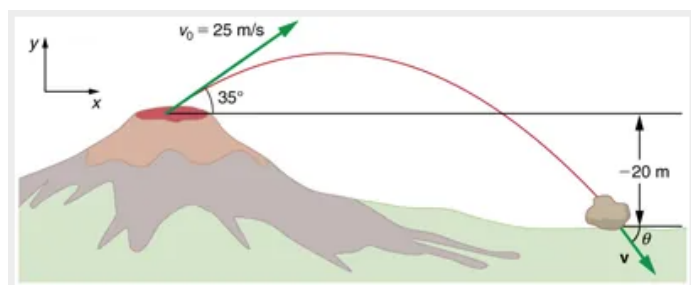


Figure 2: A volcano ejects a projectile.

2 Graphical Analysis of Kinematics

1. Consider the motion of the runner depicted in Fig. 1. (a) What is the speed of the system after $t = 4$ seconds? (b) What is the acceleration between $t = 0$ and $t = 4$ seconds? (c) What is the speed of the runner between $t = 0$ and $t = 4$ seconds?

2. Kilauea in Hawaii is the world's most continuously active volcano. Very active volcanoes characteristically eject red-hot rocks and lava rather than smoke and ash. Suppose a large rock is ejected from the volcano with a speed of 25.0 m/s and at an angle 35 degrees above the horizontal, as shown in Fig. 2. The rock strikes the side of the volcano at an altitude 20.0 m lower than its starting point. (a) Determine the *y-component* of the initial velocity of the rock. (b) Determine the *x-component* of the initial velocity.

1 Memory Bank

1. $x(t) = \frac{1}{2}at^2 + v_{i,x}t + x_i$... Horizontal position versus time with constant acceleration.
2. $v_x(t) = at + v_{i,x}$... Horizontal velocity versus time with constant acceleration.
3. $y(t) = \frac{1}{2}at^2 + v_{i,y}t + y_i$... Vertical position versus time with constant acceleration.
4. $v_y(t) = at + v_{i,y}$... Vertical velocity versus time with constant acceleration.
5. $v_f^2 = v_i^2 + 2a\Delta x$... Kinematic equation without time.
3. Note that the vertical acceleration is $-g$. (a) Calculate the time it takes the rock to land. (b) Note that the horizontal acceleration is zero. Determine the horizontal displacement of the rock from the top of the volcano when it lands.
4. **Think conceptually:** If the initial angle of the rock where 45 degrees, rather than 35 degrees, would it land farther from the volcano, or closer to the volcano?